[redacted] run report — wieting-m647-run37

precision	engineering_mixed: conservative state double on the host, float on the device
gpu	
geometry	(in mission)
mechanism	air-5-park
atmosphere	earth
closure	mixture (NASA, absolute energy)
recording	complete, 13 field frames, 121 diagnostics samples

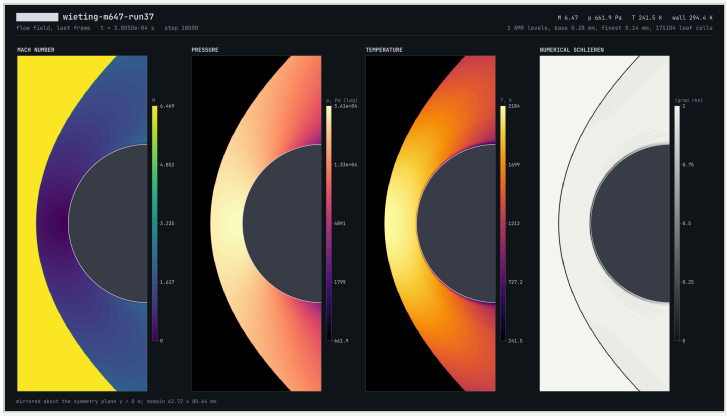
solver settings

amr_levels	2
cfl	0.600000
chemistry	on
executor	amr
flux	slau2
molecular_transport	chapman_enskog, stefan_maxwell
thermal_model	two_temperature
time_integrator	ssp_rk3
wall_thermal	isothermal
wall_velocity	no_slip

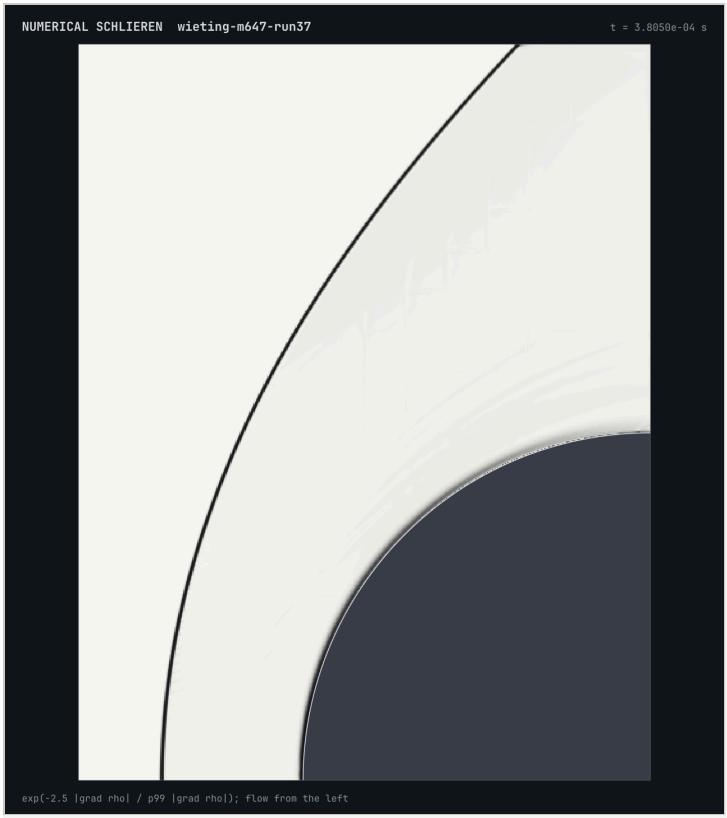
last sample

step	18000, t = 0.000380496 s, dt = 2.167e-08 s
cells	167936 (invalid 0, positivity fallbacks 0)
wall force	(0.26856, -0.15093, -1.0152e-10) N
wall heat	6.0085 W over 1.6758e-05 m2
chemistry	0 reacting cells, 0 failures, 0 s chemistry / 0 s flow

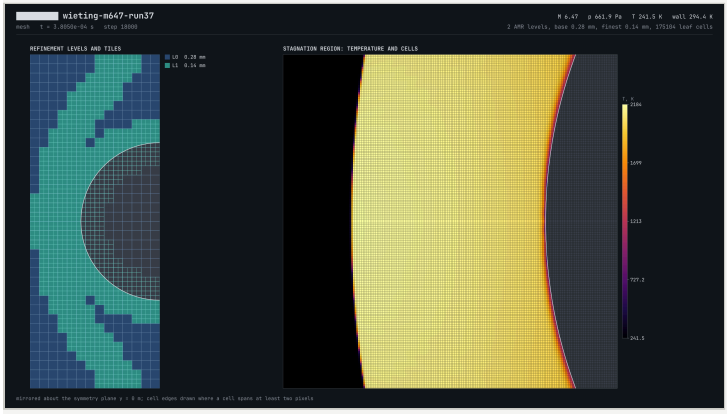
flow field



Mach number, pressure, temperature and numerical schlieren of the last frame



Numerical schlieren, upper half (as photographed in the tunnel)



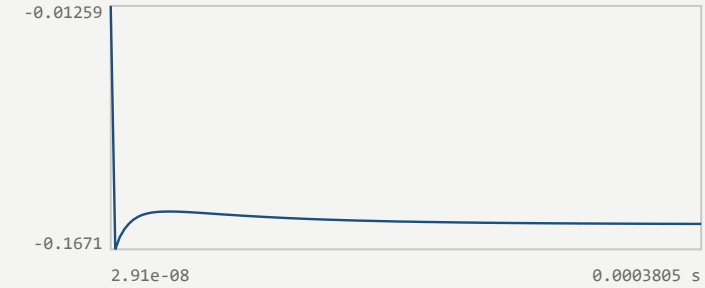
Refinement levels with tile outlines, and the stagnation region with the cells of every level

history

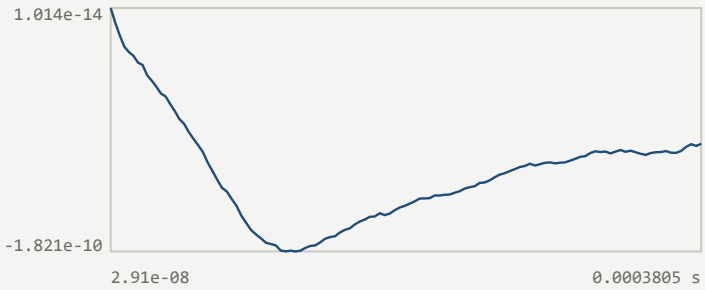
wall force x, N



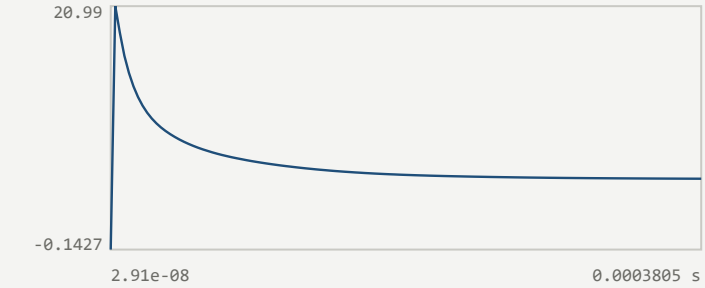
wall force y, N



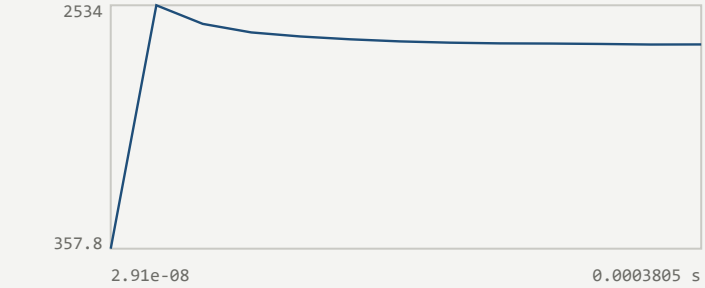
wall force z, N



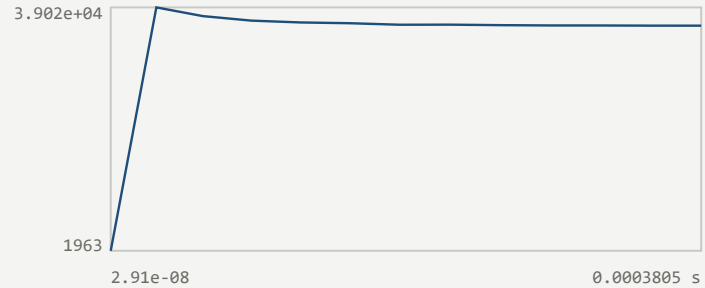
wall heat, W



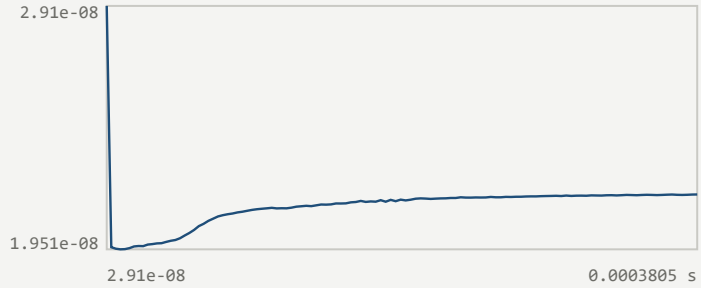
max T, K (field frames)



max p, Pa (field frames)



time step, s (CFL set, not measured)



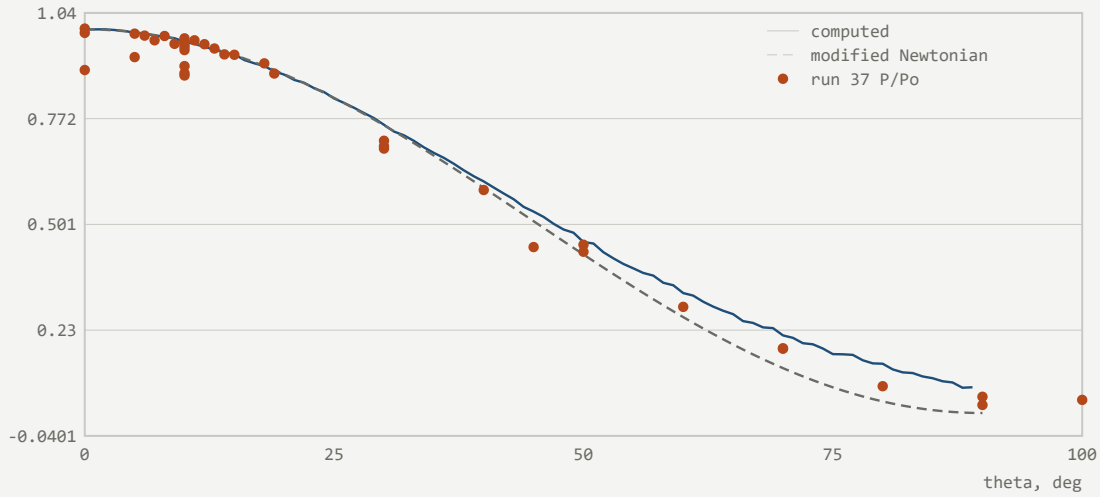
validation — Wieting 1987, run 37: cylinder in Mach 6.47 flow

case file	wieting_m647.json
reference	A. R. Wieting, NASA TM-100484 (1987), 8-ft HTT run 37: pp. 49-50 (test conditions), Figs. 17-18 (pp. 60-61), appendix C (Fay-Riddell for 2D), appendix E p. 207 (Q/Qo, P/Po)
freestream	M 6.47, p 661.9 Pa, T 241.5 K, V 2015.6 m/s, T0 2263.4 K (perfect gas, gamma 1.4)
body	cylinder R 38.1 mm, wall 294.4 K
frame	last, t = 0.000380496 s, grid step 0.14 mm (R/272)
wall heat flux	545 wall elements (amr_cut_cells, t 0.0003804960417 s)
settling	-0.37 % over the last 10 % of the run
note	run 37 table (p. 207): Qo = 59 Btu/ft2-s at a 530 R wall, Po = 5.5 psia
note	P/Po: six orifices more than 0.2 below modified Newtonian (-35, -25, -20, -17, -16 and one of -10 deg) are left out, as in Fig. 18a
note	experimental Q/Qo at theta = 0 scatters 0.95-1.04 between gauges
note	theory references assume a calorically perfect gas (gamma 1.4)

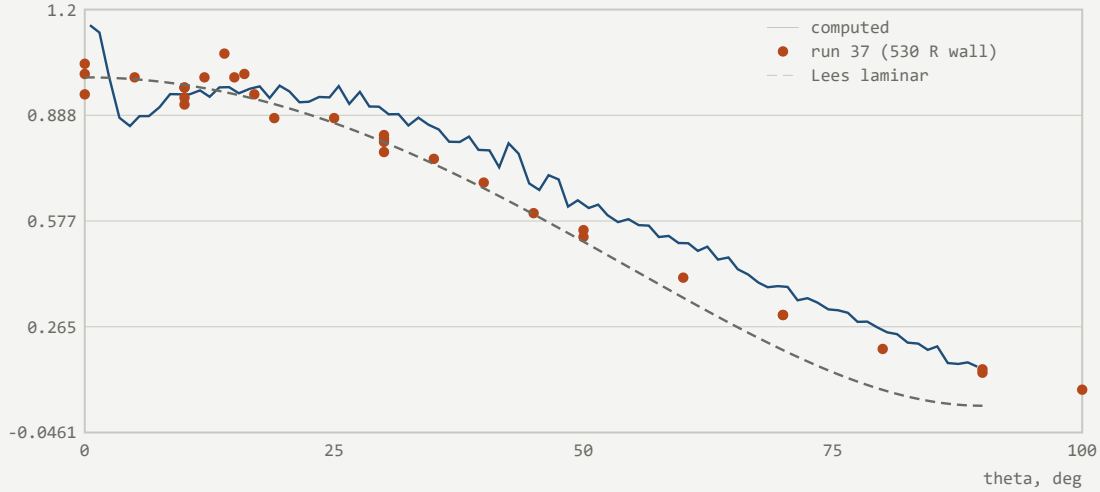
metric	computed	reference	deviation	tolerance	verdict	source
p_pitot	36.119 kPa	36.06 kPa	+0.17 %	3 %	PASS	TM-100484 pp. 49-50: pitot 5.23 +- 0.1 psia
p_pitot_theory	36.119 kPa	35.982 kPa (rayleigh_pitot)	+0.38 %	2 %	PASS	Rayleigh pitot formula, gamma 1.4
standoff	15.508 mm	16.442 mm (billig)	-5.68 %	8 %	PASS	Billig, J. Spacecraft Rockets 4 (1967) 822: 0.386 exp(4.67/M^2) R
q_stag	543.08 kW/m2	670.04 kW/m2	-18.95 %	15 %	FAIL	TM-100484 app. E: Qo = 59 Btu/ft2-s, 530 R wall
q_stag_fay_riddell	543.08 kW/m2	491.79 kW/m2 (fay_riddell)	+10.43 %	20 %	PASS	Fay & Riddell 1958, 2D constant 0.57 (TM-100484 app. C)
p_newtonian	7 angles	modified_newtonian	rms 0.027, max 0.061	0.04	PASS	Lees 1955 modified Newtonian, cos^2 + p_inf/p0 sin^2
p_run37	36 angles	points	rms 0.040, max 0.104	0.05	PASS	TM-100484 app. E p. 207, run 37
q_run37	30 angles	points	rms 0.086, max 0.204	0.08	FAIL	TM-100484 app. E, run 37, cold-wall column
q_lees	9 angles	lees	rms 0.130, max 0.187	0.08	FAIL	Lees, Jet Propulsion 26 (1956) 259

6 passed, 3 failed

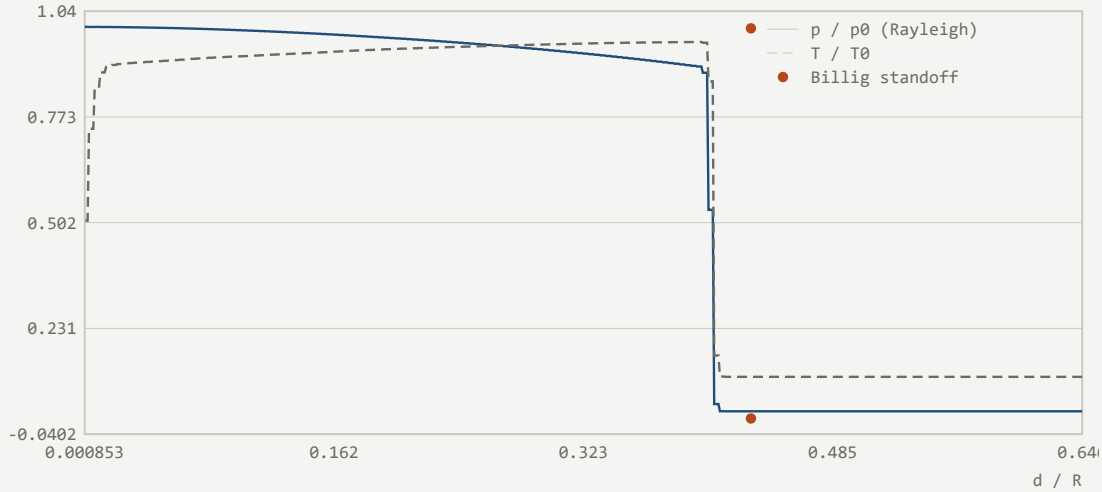
p / p0 along the body



q_w / q0 along the body



stagnation line, distance from the wall / R

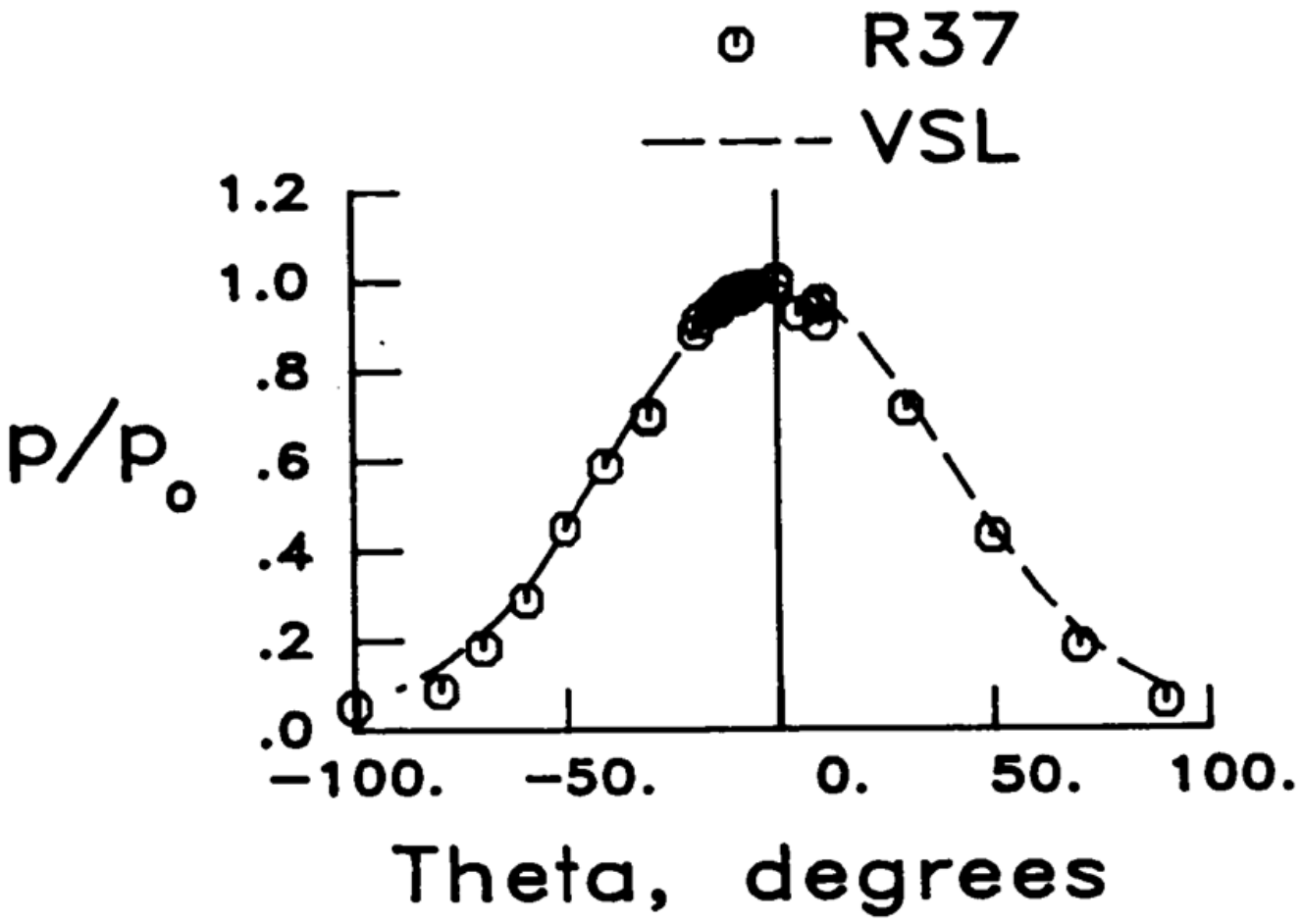


reference figures

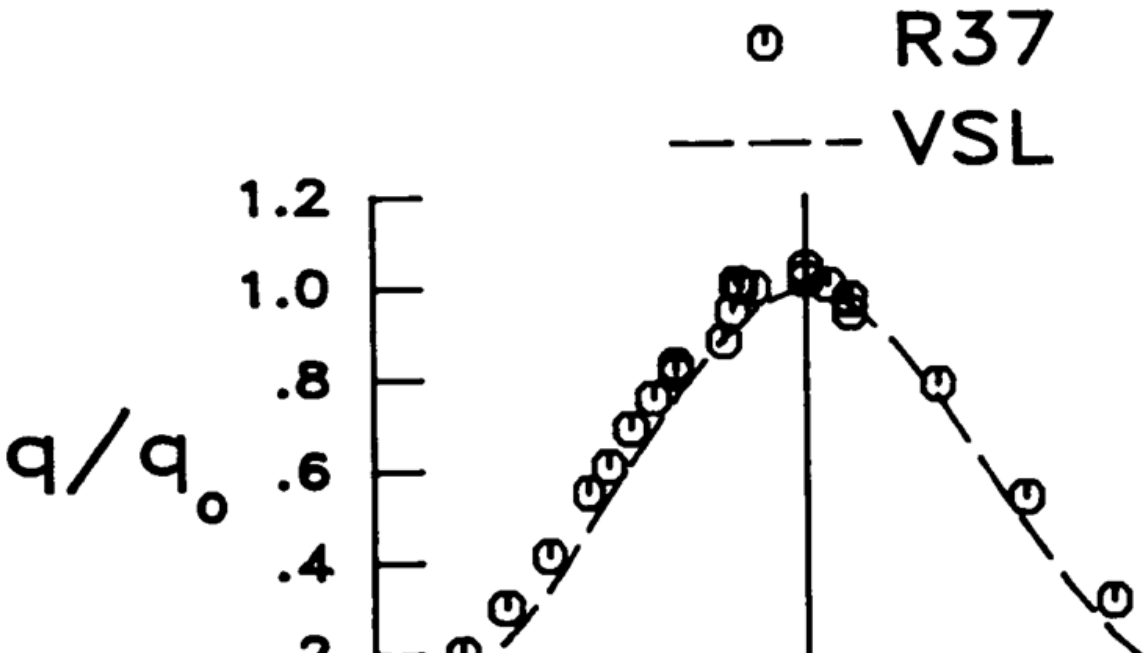
6.47 ± 0.05, pitot pressure was 5.23 ± 0.1 psia, static pressure was 0.096 ± 0.001 psia and stagnation temperature was 3400 °R ± 100 °R.

Test conditions of the M 6.47 series — TM-100484 p. 50

computed inflow: M 6.47, p 661.9 Pa, T 241.5 K; computed pitot 36.119 kPa (5.239 psia)



a) Pressure distribution



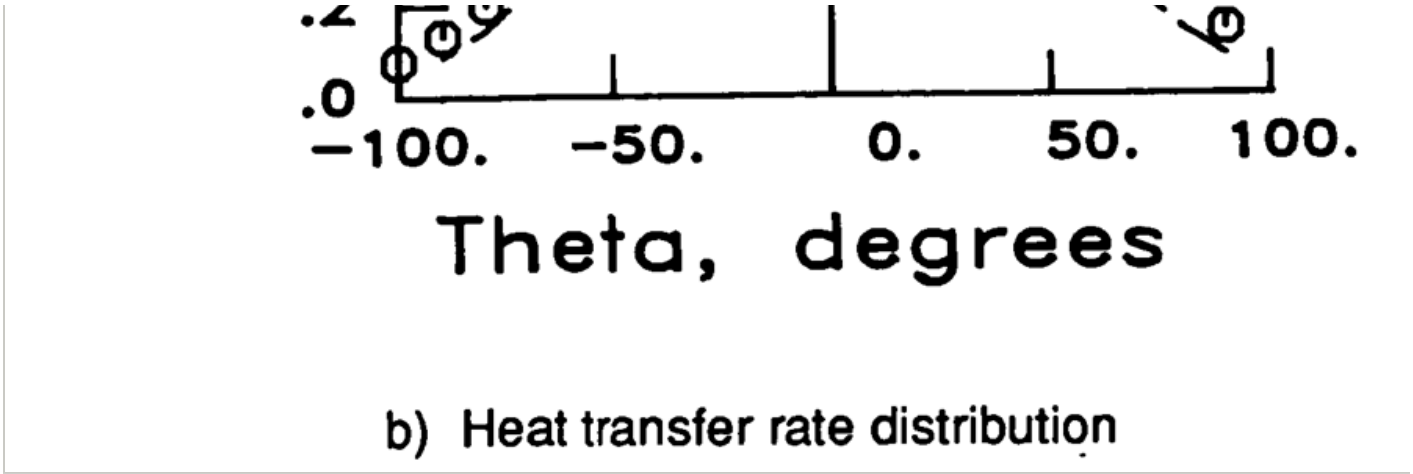
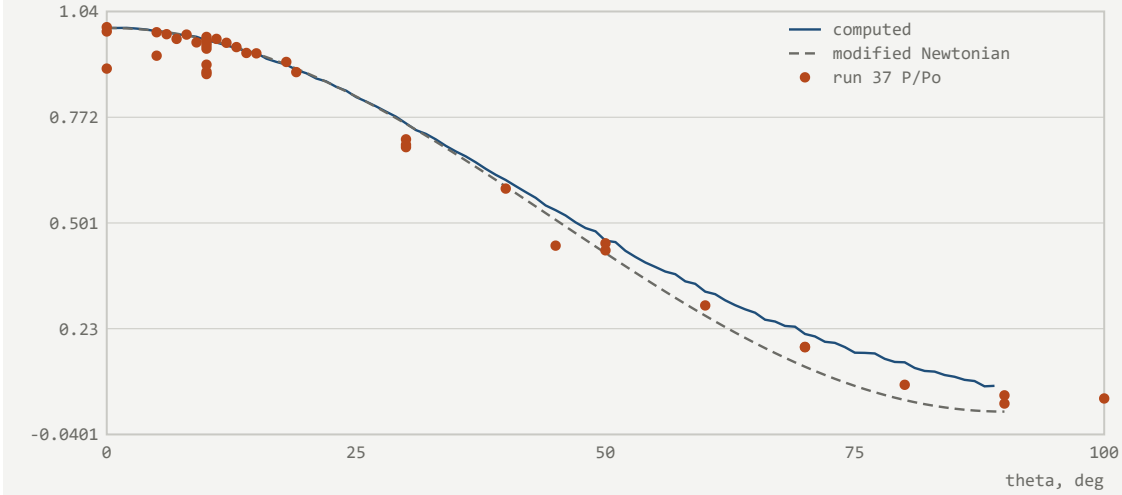


Fig. 18: p/Po and q/qo on the cylinder, run 37, with the VSL prediction — TM-100484 p. 61

computed, same quantities

p / p0 along the body



q_w / q0 along the body

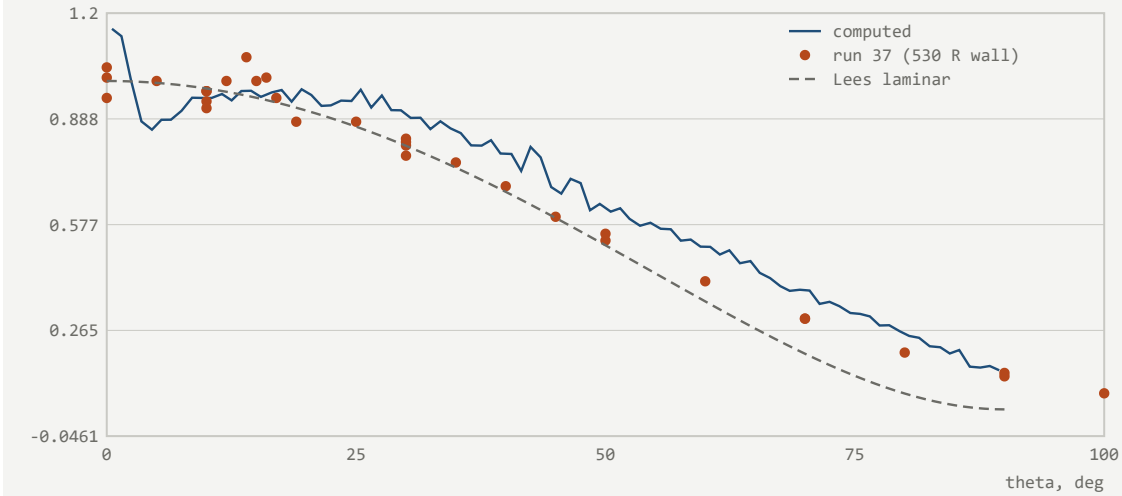
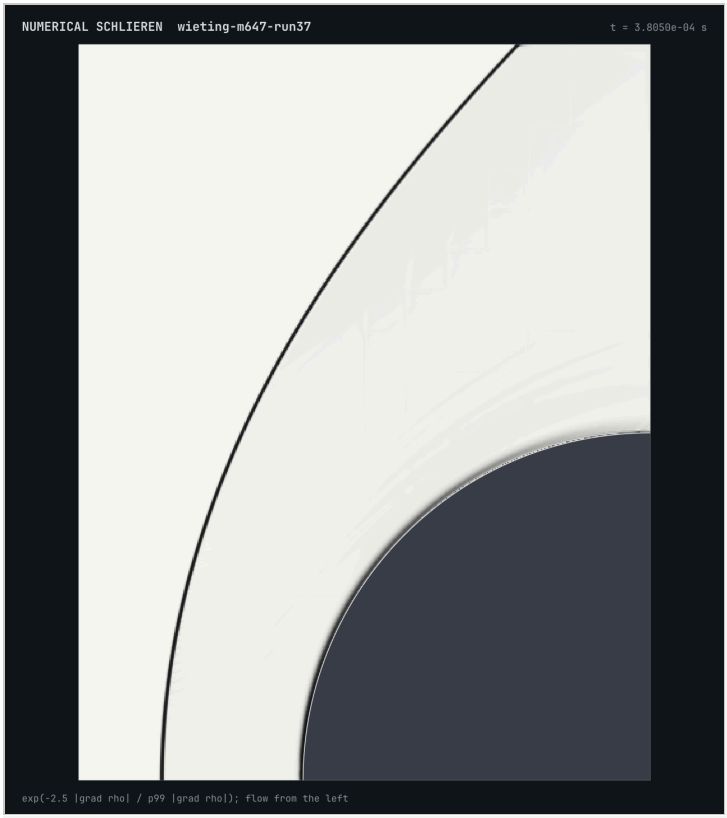




Fig. 17: schlieren photograph of the undisturbed flow, run 37 — TM-100484 p. 60



computed numerical schlieren, last frame

RUN 37

				Heat transfer		Pressure
				$Q_o = 59$ Btu/ft^2-s		$P_o = 5.5$ psia
Gage	LaRC Desig.	Theta degree	T_w °R	$Q/Q_{o[hw]}$	$Q/Q_{o[cw]}$	P/P_o
1	28	-100.0	561.	0.08	0.08	0.052
2	27	-90.0	580.	0.13	0.13	0.039
3	26	-80.0	605.	0.19	0.20	0.087
4	25	-70.0	639.	0.29	0.30	0.183
5	24	-60.0	674.	0.39	0.41	0.290
6	23	-50.0	721.	0.51	0.55	0.449
7	22	-45.0	735.	0.56	0.60	0.443
8	21	-40.0	761.	0.63	0.69	0.589
9	20	-35.0	778.	0.69	0.76	0.433
10	19	-30.0	797.	0.74	0.82	-----
11	41	-30.0	806.	0.74	0.83	0.695
12	42	-30.0	796.	0.73	0.81	0.701
13	18	-25.0	818.	0.78	0.88	0.528
14	17	-20.0	----	----	----	0.572
15	16	-19.0	816.	0.79	0.88	0.887
16	15	-18.0	----	----	----	0.913
17	14	-17.0	843.	0.84	0.95	0.628
18	13	-16.0	865.	0.88	1.01	0.630
19	12	-15.0	845.	0.88	1.00	0.935
20	11	-14.0	861.	0.94	1.07	0.936
21	10	-13.0	----	----	----	0.951
22	9	-12.0	845.	0.88	1.00	0.962
23	8	-11.0	----	----	----	0.972
24	7	-10.0	----	----	----	0.962
25	40	-10.0	----	----	----	0.977
26	43	-10.0	858.	0.81	0.92	0.888
27	44	-10.0	----	----	----	0.968
28	45	-10.0	----	----	----	0.969
29	46	-10.0	----	----	----	0.751
30	6	-9.0	----	----	----	0.963
31	5	-8.0	----	----	----	0.983
32	4	-7.0	----	----	----	0.972
33	3	-6.0	----	----	----	0.984
34	2	-5.0	----	----	----	0.989
35	1	0.0	849.	0.89	1.01	1.002
36	38	0.0	837.	0.84	0.95	0.896
37	39	0.0	858.	0.91	1.04	0.991
38	29	5.0	845.	0.88	1.00	0.929
39	30	10.0	839.	0.86	0.97	0.906
40	35	10.0	839.	0.86	0.97	0.955
41	36	10.0	833.	0.83	0.94	0.882
42	37	10.0	----	----	----	0.947
43	31	30.0	784.	0.71	0.78	0.715
44	32	50.0	716.	0.50	0.53	0.431
45	33	70.0	643.	0.29	0.30	0.194

45	33	70.0	643.	0.29	0.30	0.184
46	34	90.0	586.	0.14	0.14	0.060

Run 37 gauge table: theta, wall temperature, Q/Qo (hot and 530 R wall), P/Po — TM-100484 appendix E, p. 207

computed, the metrics above

metric	computed	reference	deviation	tolerance	verdict	source
p_pitot	36.119 kPa	36.06 kPa	+0.17 %	3 %	PASS	TM-100484 pp. 49-50: pitot 5.23 +- 0.1 psia
p_pitot_theory	36.119 kPa	35.982 kPa (rayleigh_pitot)	+0.38 %	2 %	PASS	Rayleigh pitot formula, gamma 1.4
standoff	15.508 mm	16.442 mm (billig)	-5.68 %	8 %	PASS	Billig, J. Spacecraft Rockets 4 (1967) 822: 0.386 exp(4.67/M^2) R
q_stag	543.08 kW/m2	670.04 kW/m2	-18.95 %	15 %	FAIL	TM-100484 app. E: Qo = 59 Btu/ft2-s, 530 R wall
q_stag_fay_riddell	543.08 kW/m2	491.79 kW/m2 (fay_riddell)	+10.43 %	20 %	PASS	Fay & Riddell 1958, 2D constant 0.57 (TM-100484 app. C)
p_newtonian	7 angles	modified_newtonian	rms 0.027, max 0.061	0.04	PASS	Lees 1955 modified Newtonian, cos^2 + p_inf/p0 sin^2
p_run37	36 angles	points	rms 0.040, max 0.104	0.05	PASS	TM-100484 app. E p. 207, run 37
q_run37	30 angles	points	rms 0.086, max 0.204	0.08	FAIL	TM-100484 app. E, run 37, cold-wall column
q_lees	9 angles	lees	rms 0.130, max 0.187	0.08	FAIL	Lees, Jet Propulsion 26 (1956) 259

separate files

pdf report.pdf
png ref_fig17_schlieren_run37.png
png ref_fig18_distributions_run37.png
png ref_table_run37_appendix_e.png
png ref_test_conditions_p50.png
png report_fields.png
png report_mesh.png
png report_p_distribution.png
png report_q_distribution.png
png report_schlieren.png
png report_stagnation_line.png